

Agentic Harness Router: Obstacles Update

Changes since the 29 Sep version on GitHub | Arsh Mansuri | 3 Oct 2026

Summary

The 29 Sep version on GitHub is still mostly accurate. Since then: (1) we still have no router that beats always using the best single harness; (2) several setup problems cost time; and (3) a full study is now designed but not yet run, which pushes back the main result. Setup is unchanged: GPT-5.6-Luna runs every task, and the router only chooses the harness (terminus-2, mini-swe-agent or pi).

1. No router beats the best single harness yet

- Harness choice does matter per task: on 21 tasks, perfect picking would solve about 16, compared with 13 for always mini-swe-agent. Every router we have built scores about 13 or lower.
- The three harnesses give the same result on 13 of the 21 tasks, and 4 tasks are never solved, so there are few tasks where a good pick can help.
- Results vary between repeat runs: one task passed twice on mini-swe-agent and then failed on it. With mostly one run per task and harness, single comparisons are not reliable.
- The 7 held-out test tasks have been looked at several times, so they are no longer a clean test.

2. Setup and operational problems

Problem	Effect	Status
iCloud had offloaded parts of the project's Python environment	Router startup took about 20 minutes	Partly fixed; folder needs "Keep Downloaded"
OpenRouter key hit its credit limit	Runs failed with credit errors	Fixed (limit raised to \$20)
Overnight batch run killed by the editor shortly after starting	No results from that night	Fixed: run from a normal Terminal window

3. Timeline impact

To get a trustworthy answer, I have written a study plan (STUDY_PLAN.md) with success criteria set before any data is collected:

- 45 tasks: 27 for development and 18 for testing that have never been run before, so nobody has seen their results.
- Every task on every harness, 3 runs each: 405 runs, about 11 to 20 hours and \$5 to \$12.
- Routing counts as an improvement if either: the pass rate is at least 5 points higher than the best single harness, or cost per pass is at least 30% lower without losing more than 3 points of pass rate.

The scripts are ready, but the big run has not happened yet because of the problems above. The main result is therefore about a week behind. The study plan will be pushed to GitHub.